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Answer

TID	Items
1	Bread, Milk, Chips, Mustard
2	Beer, Diaper, Bread, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk, Chips
5	Coke, Bread, Diaper, Milk
6	Beer, Bread, Diaper, Milk, Mustard
7	Coke, Bread, Diaper, Milk

Minimum Support Calculation:

$$\text{Minimum support} = 40\% = 0.40 \times 7 = 2.8$$

Since support count must be an integer, an itemset is frequent if it appears in at least 3 transactions (we round up to the nearest integer threshold).

The Apriori algorithm proceeds level-wise: Start with frequent 1-itemsets, then generate candidates for larger itemsets by joining previous frequent sets, and prune those below the support threshold.

Step 1: Generate Frequent 1-Itemsets (L_1)

Scan the dataset to count the support for each individual item. Prune items with support < 3 .

- Milk: Appears in TID 1, 3, 4, 5, 6, 7 $\rightarrow$ Support = 6
- Bread: Appears in TID 1, 2, 4, 5, 6, 7 $\rightarrow$ Support = 6
- Diaper: Appears in TID 2, 3, 4, 5, 6, 7 $\rightarrow$ Support = 6
- Beer: Appears in TID 2, 3, 4, 6 $\rightarrow$ Support = 4
- Coke: Appears in TID 3, 5, 7 $\rightarrow$ Support = 3
- Chips: Appears in TID 1, 4 $\rightarrow$ Support = 2 (pruned)
- Mustard: Appears in TID 1, 6 $\rightarrow$ Support = 2 (pruned)
- Eggs: Appears in TID 2 $\rightarrow$ Support = 1 (pruned)

Frequent 1-Itemsets:

- {Bread} (6)
- {Beer} (4)
- {Coke} (3)
- {Diaper} (6)
- {Milk} (6)

Step 2: Generate Frequent 2-Itemsets (L_2)

Generate candidate 2-itemsets by joining all pairs from L_1 . Scan the dataset to count support for each pair. Prune pairs with support < 3 .

- {Bread, Milk}: Appears in TID 1, 4, 5, 6, 7 $\rightarrow$ Support = 5
- {Bread, Diaper}: Appears in TID 2, 4, 5, 6, 7 $\rightarrow$ Support = 5

- {Milk, Diaper}: Appears in TID 3, 4, 5, 6, 7 → Support = 5
- {Beer, Diaper}: Appears in TID 2, 3, 4, 6 → Support = 4
- {Bread, Beer}: Appears in TID 2, 4, 6 → Support = 3
- {Beer, Milk}: Appears in TID 3, 4, 6 → Support = 3
- {Coke, Diaper}: Appears in TID 3, 5, 7 → Support = 3
- {Coke, Milk}: Appears in TID 3, 5, 7 → Support = 3

(Other pairs, e.g., {Bread, Coke}, {Beer, Coke}, have support < 3 and are pruned.)

Frequent 2-Itemsets:

- {Bread, Milk} (5)
- {Bread, Diaper} (5)
- {Milk, Diaper} (5)
- {Beer, Diaper} (4)
- {Bread, Beer} (3)
- {Beer, Milk} (3)
- {Coke, Diaper} (3)
- {Coke, Milk} (3)

Step 3: Generate Frequent 3-Itemsets (L_3)

Generate candidate 3-itemsets by joining pairs from L_2 that share two items (e.g., {Bread, Milk} + {Bread, Diaper} → {Bread, Milk, Diaper}). Scan the dataset to count support. Prune triplets with support < 3.

- {Bread, Milk, Diaper}: Appears in TID 4, 5, 6, 7 → Support = 4
- {Beer, Bread, Diaper}: Appears in TID 2, 4, 6 → Support = 3
- {Beer, Milk, Diaper}: Appears in TID 3, 4, 6 → Support = 3
- {Coke, Milk, Diaper}: Appears in TID 3, 5, 7 → Support = 3

(Other candidates, e.g., {Bread, Beer, Milk}, {Bread, Coke, Diaper}, have support < 3 and are pruned.)

Frequent 3-Itemsets:

- {Bread, Milk, Diaper} (4)
- {Beer, Bread, Diaper} (3)
- {Beer, Milk, Diaper} (3)
- {Coke, Milk, Diaper} (3)

Step 4: Check for Larger Itemsets (4-Itemsets and Beyond)

Generate candidate 4-itemsets by joining triplets from L_3 (e.g., {Bread, Milk, Diaper} + {Beer, Milk, Diaper} → {Beer, Bread, Milk, Diaper}). Scan the dataset.

- {Beer, Bread, Milk, Diaper}: Appears in TID 4, 6 → Support = 2 (pruned)
- {Coke, Bread, Milk, Diaper}: Appears in TID 5, 7 → Support = 2 (pruned)
- {Beer, Coke, Milk, Diaper}: Appears in TID 3 → Support = 1 (pruned)

No 4-itemsets (or larger) have support ≥ 3 . The process stops here.

Final List of All Frequent Itemsets

1-Itemsets:

- {Bread} (6)
- {Beer} (4)
- {Coke} (3)
- {Diaper} (6)
- {Milk} (6)

2-Itemsets:

- {Bread, Milk} (5)
- {Bread, Diaper} (5)
- {Milk, Diaper} (5)
- {Beer, Diaper} (4)
- {Bread, Beer} (3)
- {Beer, Milk} (3)
- {Coke, Diaper} (3)
- {Coke, Milk} (3)

3-Itemsets:

- {Bread, Milk, Diaper} (4)
- {Beer, Bread, Diaper} (3)
- {Beer, Milk, Diaper} (3)
- {Coke, Milk, Diaper} (3)

No frequent itemsets of size 4 or larger.

Conclusively : The strongest pattern in the data is {Bread, Milk, Diaper}, which appears together in 4 out of 7 times